Basic Principles of Medicine 1
Module : Musculoskeletal System
MSK DLA

DLA Title: Triplet Repeat Disorders & Pharmacogenetics:

Triplet Repeat Disorders

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Acknowledgements: Mary Maj, PhD, Sharmila Upadhya, MBBS, MD, Andrew K. Sobering, PhD

Learning Objectives:

SOM.MKIII.BPM1.2.MSK.1.GNET.0166	Explain how repetitive sequences show instability such that in some individuals the size of the repeat expands in the germ line
SOM.MKIII.BPM1.2.MSK.1.GNET.0167	Explain how most triple repeat disorders show decreased age of onset and increased severity when the repeat size expands and how parental origin may sometimes influence expansions and therefore anticipation
SOM.MKIII.BPM1.2.MSK.1.GNET.0168	Describe fragile X syndrome
SOM.MKIII.BPM1.2.MSK.1.GNET.0169	Explain how fragile X syndrome is a disorder caused by a loss of expression of a gene
SOM.MKIII.BPM1.2.MSK.1.GNET.0170	Describe fragile X-associated tremor ataxia syndrome and how it is caused by an overexpression of a gene
SOM.MKIII.BPM1.2.MSK.1.GNET.0171	Describe Huntington disease (HD): how it may be diagnosed, both symptomatic and presymptomatic, via a genetic test; therapeutic use of antisense oligonucleotides
SOM.MKIII.BPM1.2.MSK.1.GNET.0172	Describe the genetics and clinical features of myotonic dystrophy
SOM.MKIII.BPM1.2.MSK.1.GNET.0173	Describe the genetics and clinical features of Friedrich ataxia and spinocerebellar ataxia

Repetitive Sequence Instability & Expansion:

- Most gene mutations are stable, inheritable change in the DNA sequence.

- Repeat mutations are unstable and change over time – grow or shrink.

- Different disorders, different repeat sequence e.g. tri, tetra, hexa etc and different thresholds of expansion for disease development.

ACTCAGCAGCAGCAGCAGTTGG

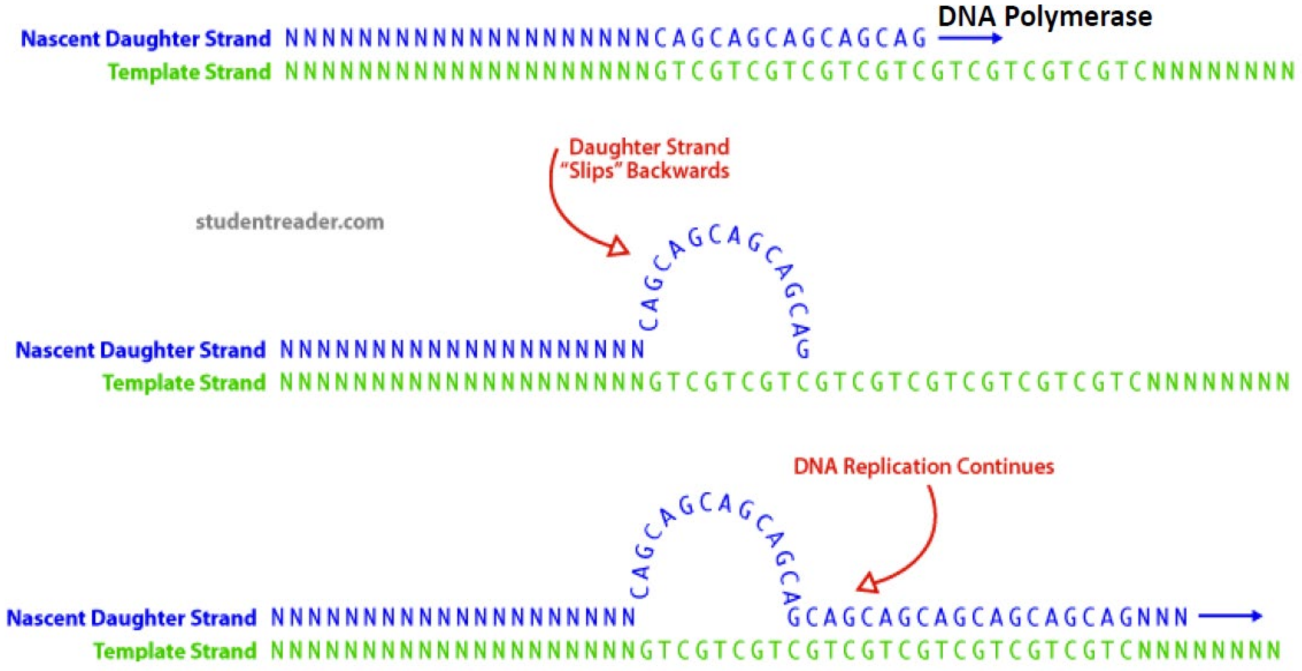
Expansion may occur as DNA is copied during mitosis or meiosis (gameteogenesis)

ACTCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGCAGTTG

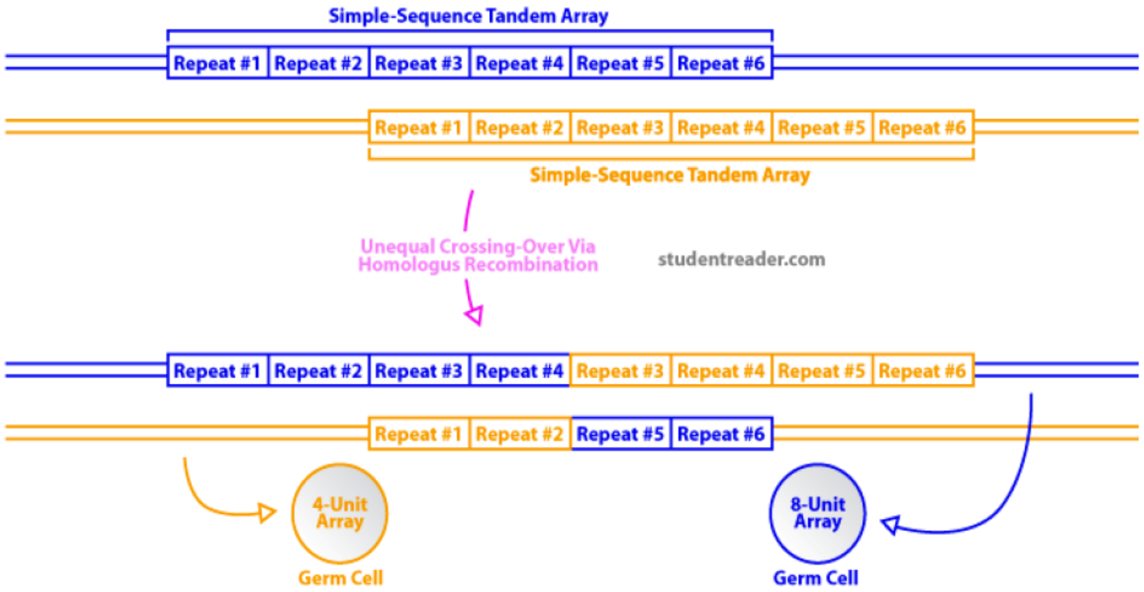
- Some disorders have a premutation state – expansion close to threshold but not over!

Mode of expansion:

- **Slippage** – DNA polymerase arrested and dissociates, new synthesized strand separates and reanneals at a different point in repeat sequence.



- **Unequal crossing over** – sister chromatids repetitive sequences do not line up exactly = one shorter, one longer.

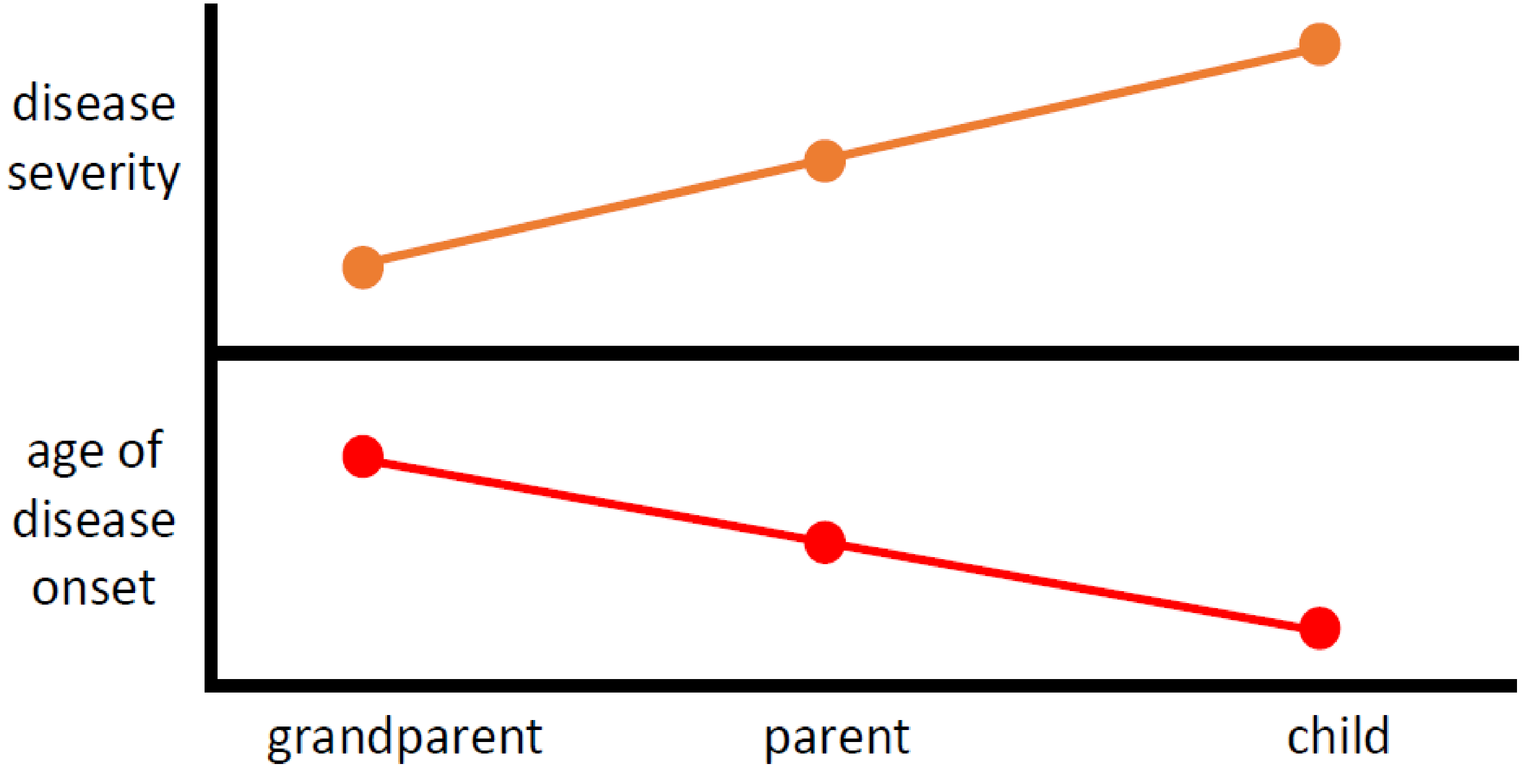


SOM.MKIII.BPM1.2.MSK.1.GNET.0166

Anticipation & Meiotic Drive:

- Expansion of the repeat sequence may occur as passed down generations.

- Result is an increase in disease severity and earlier age of onset between generations = **Anticipation.**

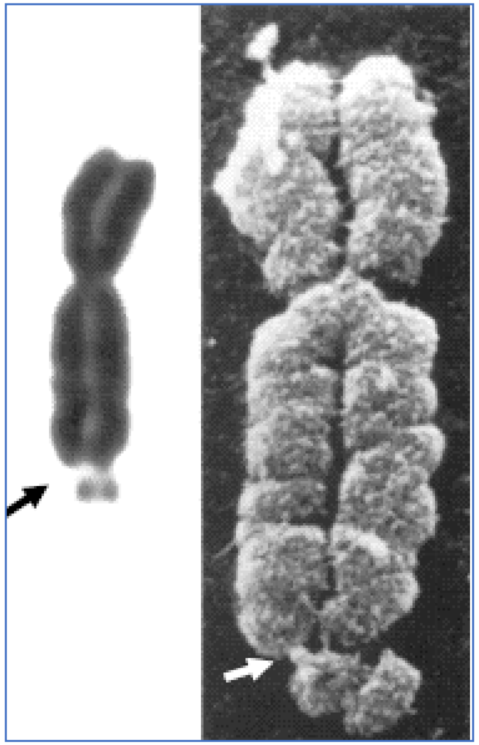


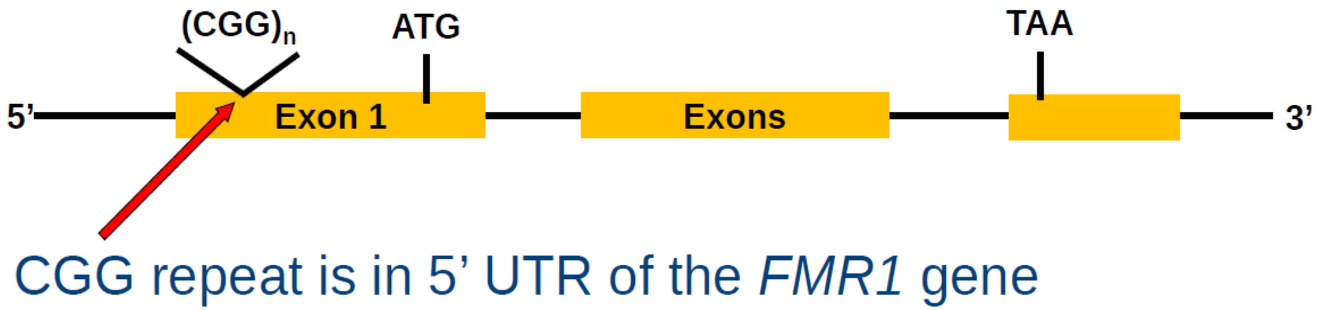
- Some disorders show sex bias – dramatic expansion of the repeat in one sex over the other i.e. during spermatogenesis or oogenesis = increase in severity more common males or females.

- Phenomenon termed **Meiotic Drive!**

Fragile X syndrome:

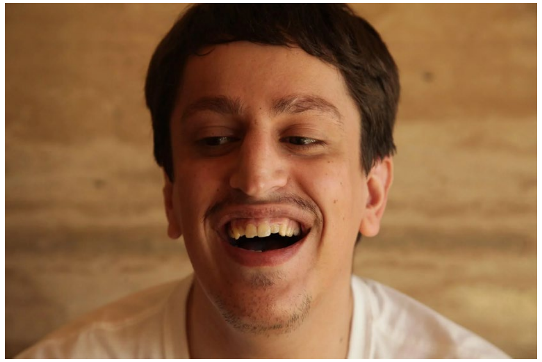
- X-linked disorder – not truly fragile, only appears in cytogenetic test, culturing in limited folate = unusual metaphase X-chr.
- CGG expansion in the 5' UTR of the *FMR1* gene > 200 repeats (normal 10-50 copies).
- Hypermethylation of expanded CpG island = gene silenced, no expression of protein.





- Diagnosis not by cytogenetic test – PCR amplification with gene-specific primers across repeat region and determination of size by gel electrophoresis.

Fragile X syndrome:



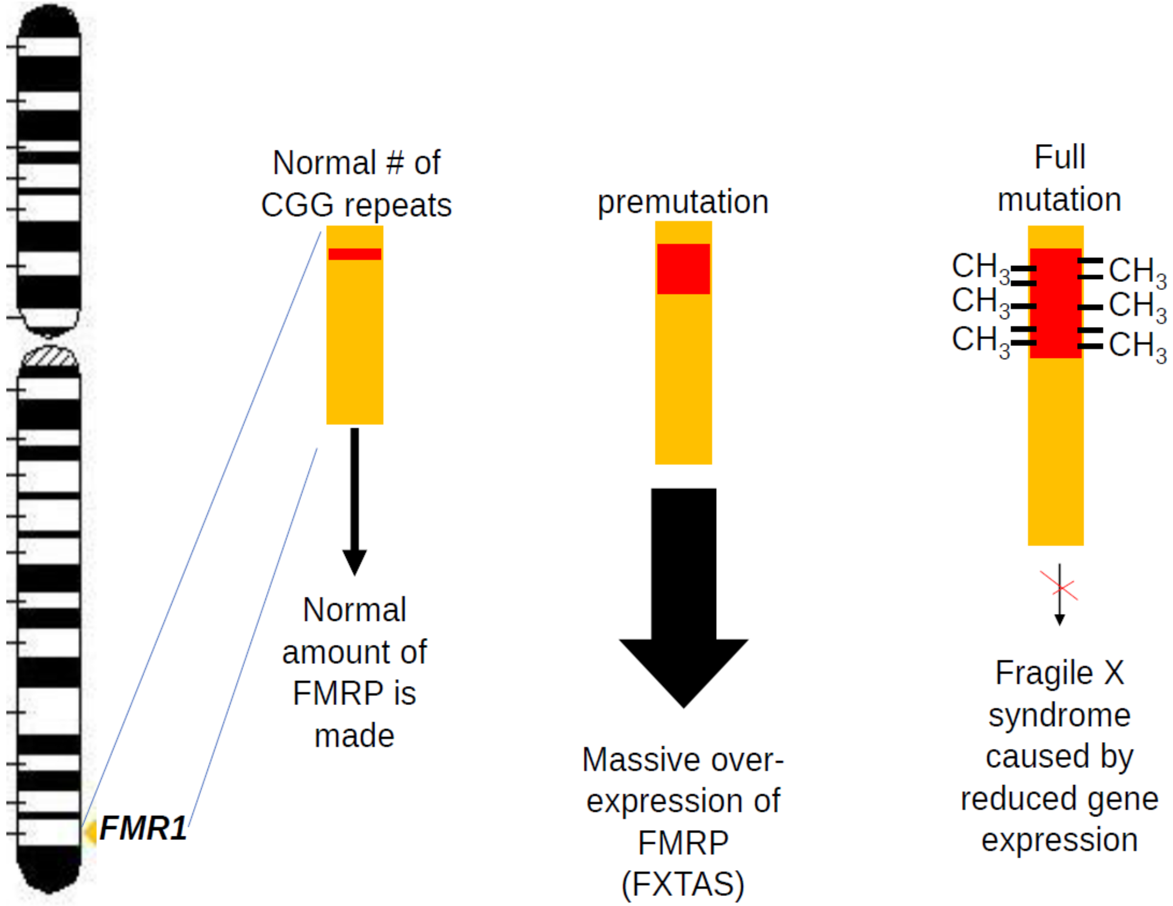
Clinical Features:

- Improper development of neuronal synapses.
- Intellectual disability and learning difficulties (cluttered, nervous speech).
- Autistic features.
- Elongated face – prominent ears.
- Reduced muscle tone and flat feet.
- Macroorchidism (enlarged testis) after puberty.

- More common and severe in males (females – variable expressivity and reduced penetrance).
- Mother with premutation normal, repeat expansion during oogenesis = son with disorder (**anticipation**).
- Some females with full mutation have abnormal phenotypes.

Fragile X-associated tremor ataxia syndrome:

- 3 important variations in *FMR1* gene.
- FXTAS – premutation = number of repeats below threshold.



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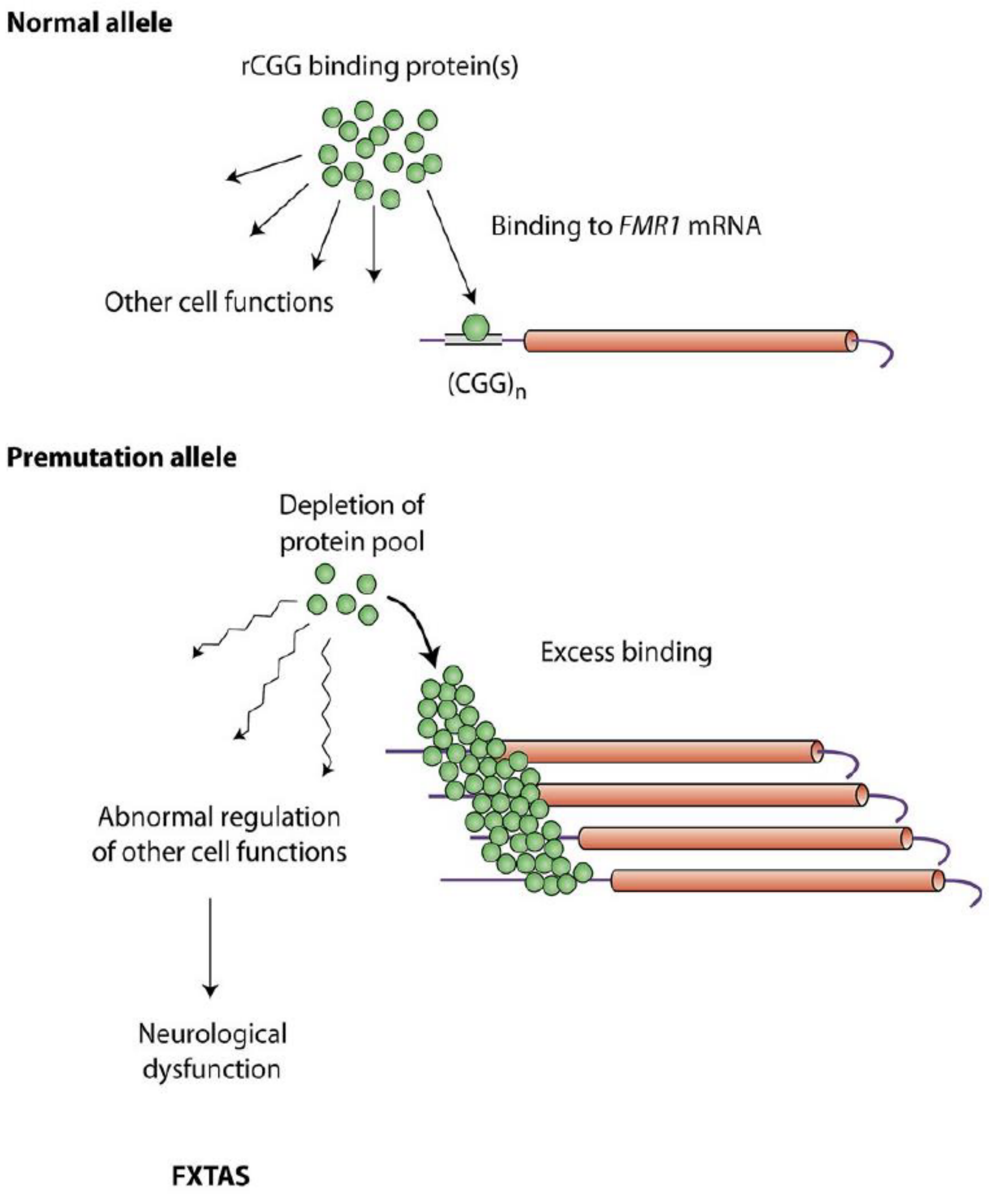


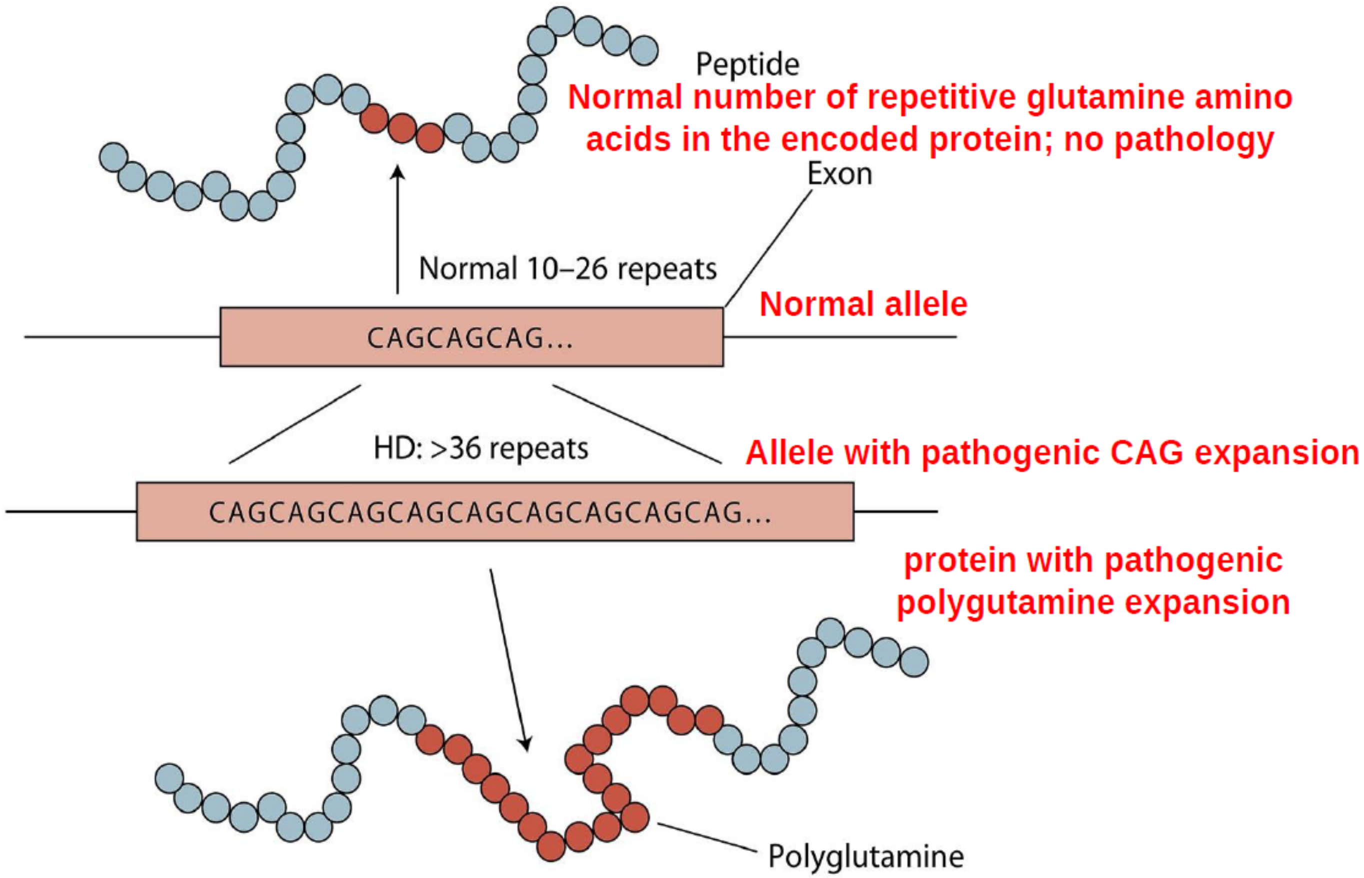
Huntington disease (HD):

- Autosomal dominantly inherited neurodegenerative disorder.
- Characterised by progressive dementia and involuntary movements.
- Symptoms caused by loss of neurons in basal ganglia and cortex.
- Mean age of onset is approx. 40 years old (early & late depending on size of repeat).
- Typical course of disease is 10 -20 years from onset:
    - Degeneration of neurons of cerebral cortex
    - Chorea due to degeneration of basal ganglia
    - Cognitive & language decline
    - Inexorable decline & death

HD molecular mechanism:

- CAG repeat in exon of *Huntington* gene - polyglutamine repeat in encoded protein (**polyglutamine tract expansion disorder**).

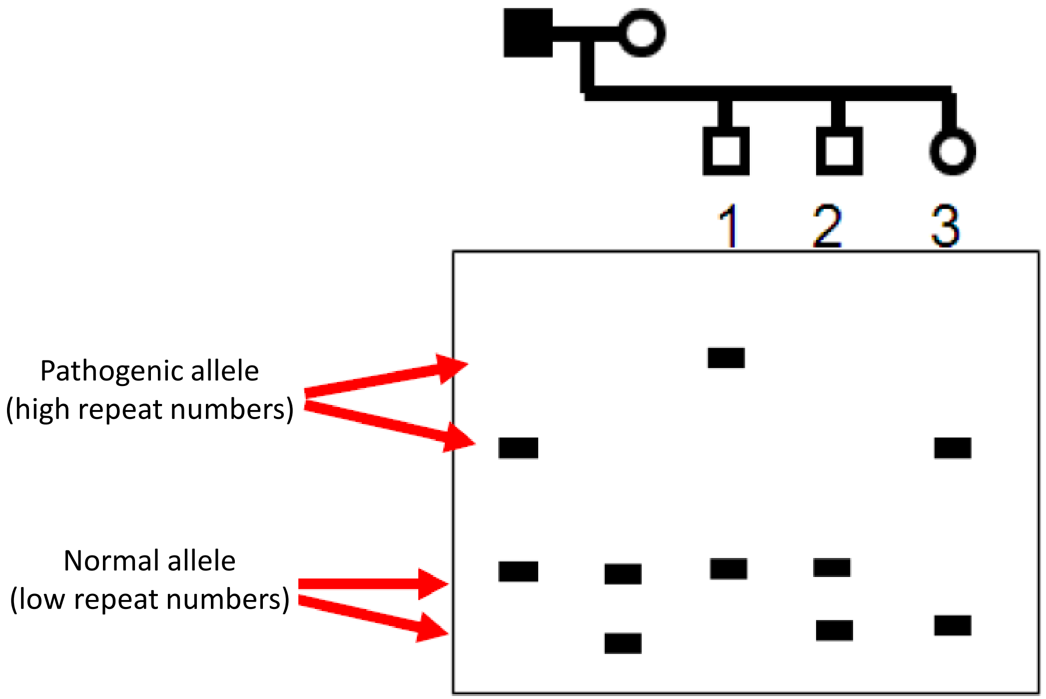
- Leads to "***gain of function***" or "***attainment of a novel function***".

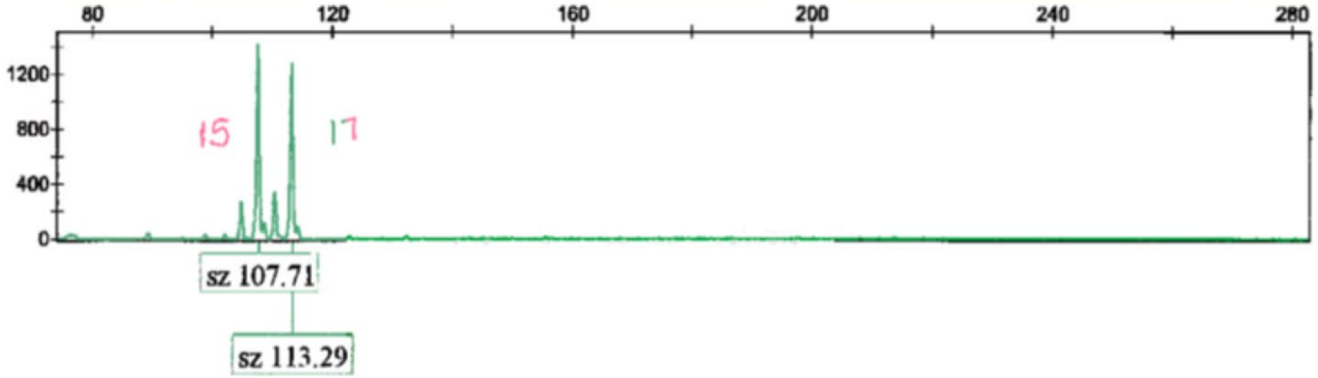


HD Diagnosis:

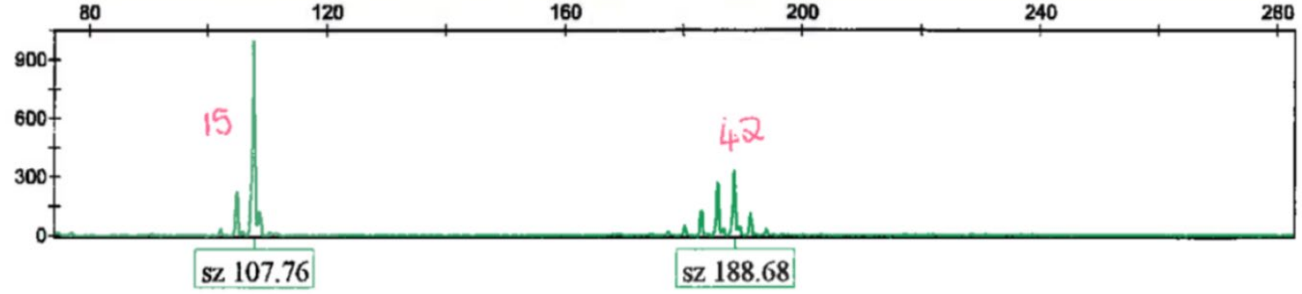
- Diagnosis, both symptomatic & pre-symptomatic (ethical issue?) via PCR amplification of repeat region.

- Analyse size of product by gel electrophoresis or column chromatography.





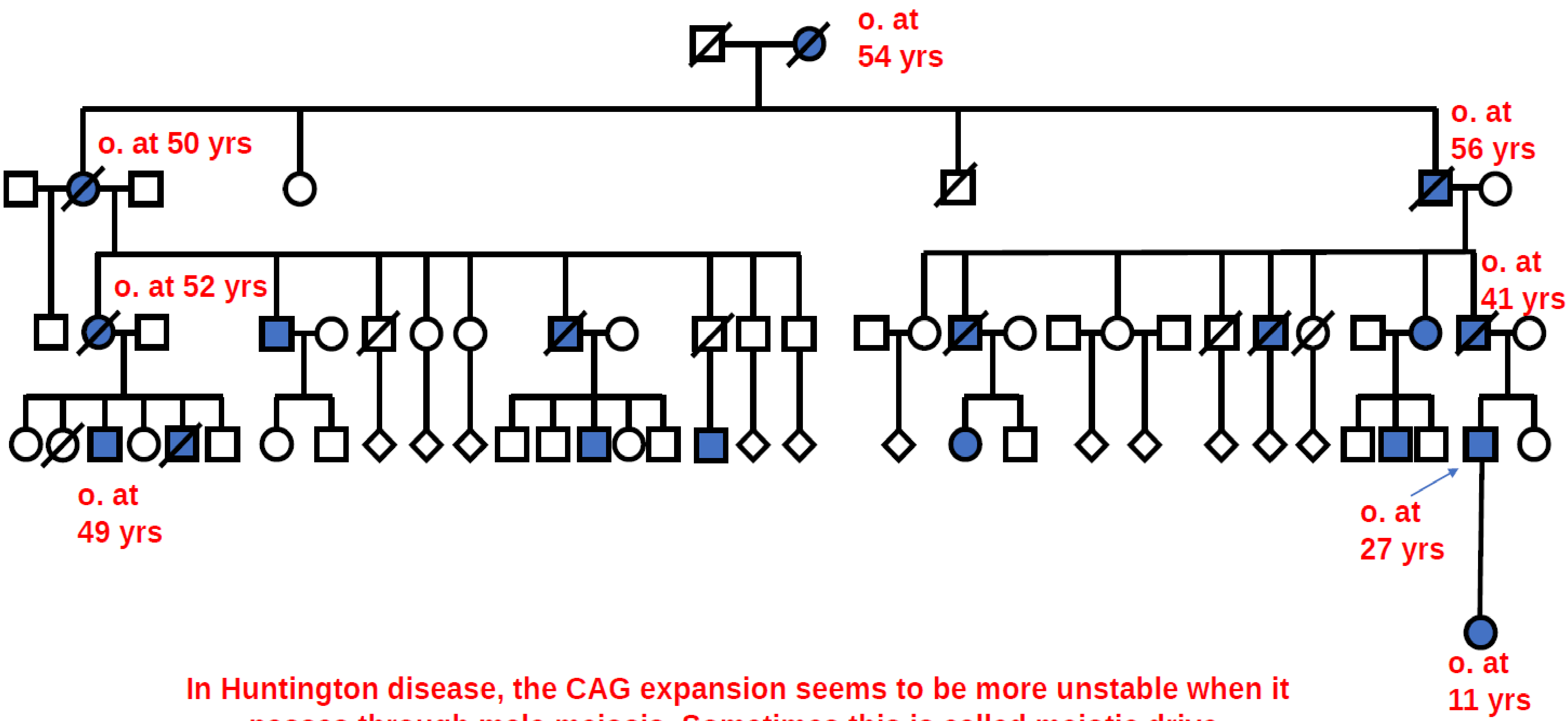
Patient who does not have Huntington disease



Patient with Huntington disease

HD Anticipation & Meiotic Drive:

- Anticipation in HD is more severe down male lineage.



In Huntington disease, the CAG expansion seems to be more unstable when it passes through male meiosis. Sometimes this is called meiotic drive.

Spinocerebellar Ataxia:

- Autosomal dominant, progressive neurodegenerative disorder.

- Displays high degree of locus heterogeneity - > 40 different genes identified in patients.

- Most cases due to triplet repeat expansion of a CAG in an exon (similar to HD).

Clinical Features:

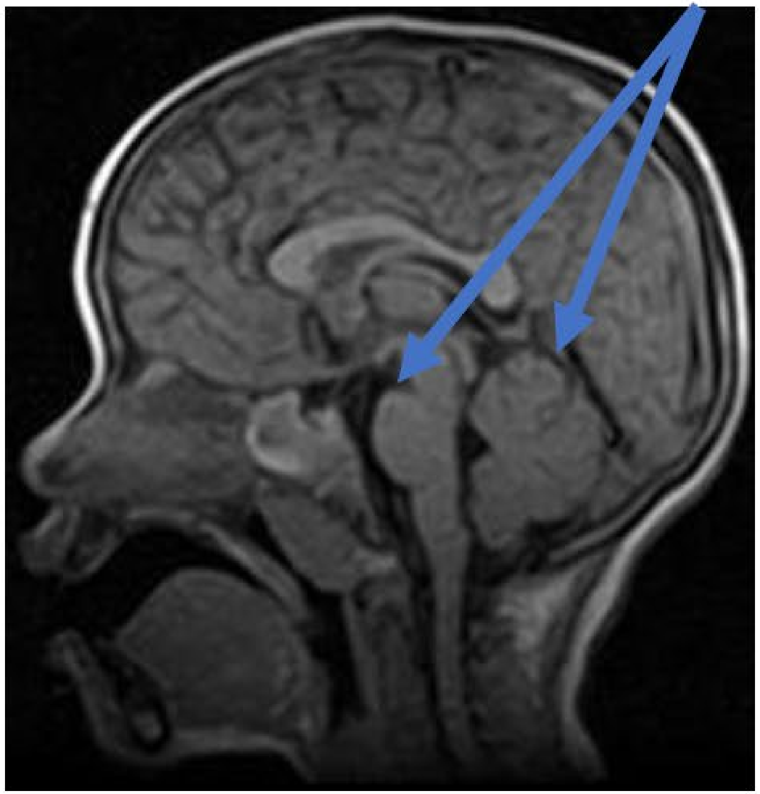
- Progressive neurodegeneration.

- Ataxia, dysphagia, dysarthria.

- Muscle weakness.

Spinocerebellar Ataxia:

Pons and cerebellar structures are normal

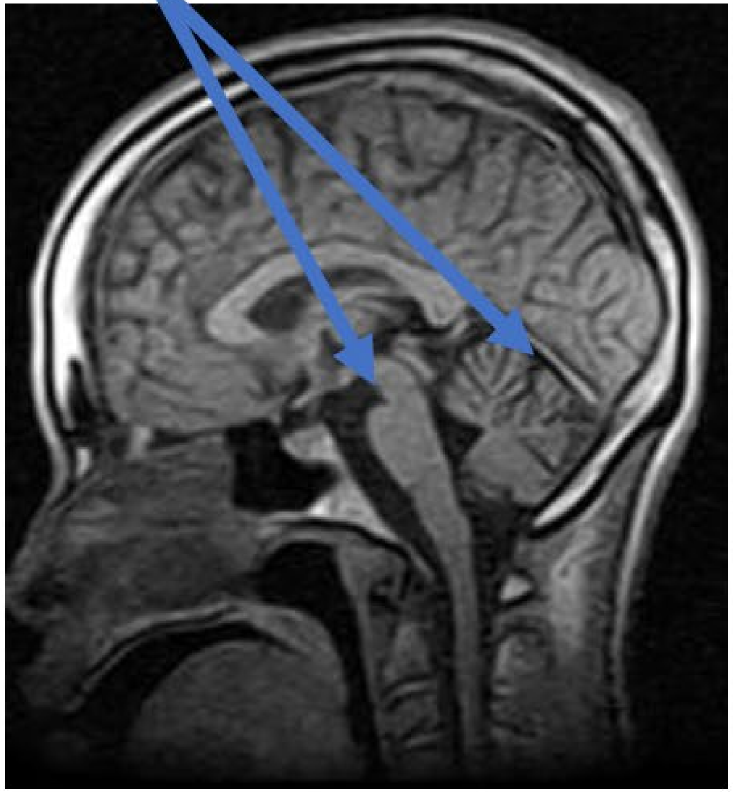


Before symptom onset

Atrophy of the cerebellum becomes apparent

Four years after symptom onset

Flattening of the pons and progressive cerebellar degeneration



Eight years after symptom onset

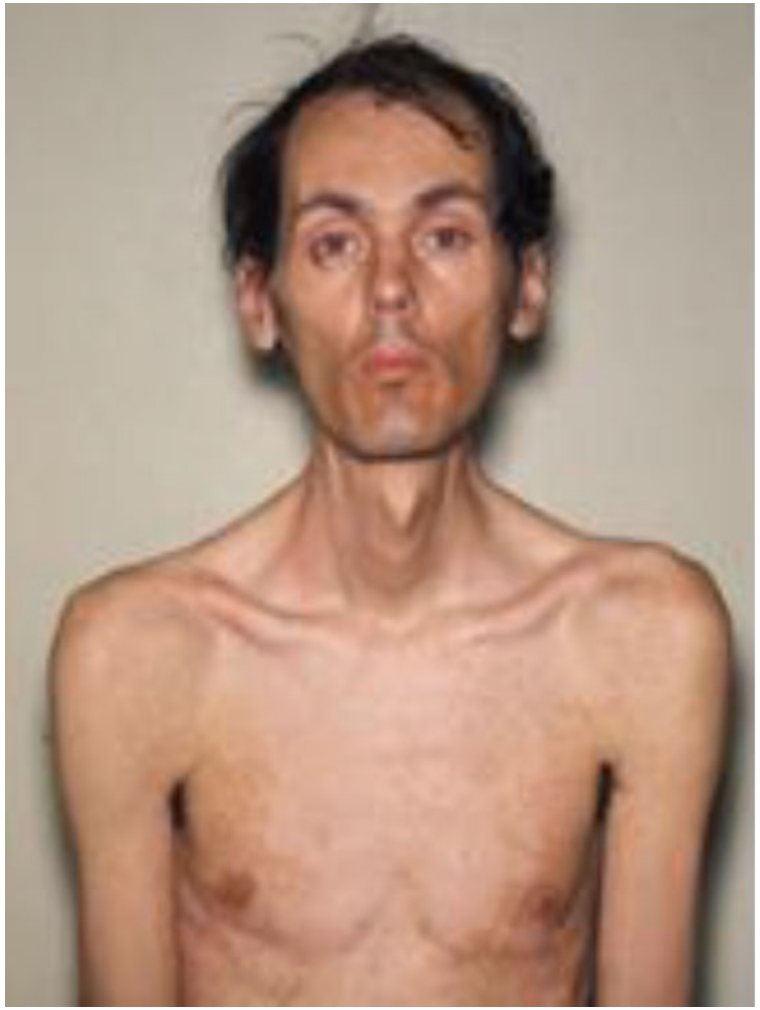
Human Genetics and Genomics: Patterns of Inheritance
DM, FA, and SCA

Myotonic Dystrophy:

- Autosomal dominant inheritance.
- Myotonus – slow or difficulty with muscle relaxation after contraction (multi- systemic disease with muscle wasting).
- Trinucleotide repeat expansion in 3' UTR of *DMPK* gene.

Clinical Features:

- Myotonia.
- Weakness in distal muscles.
- Weakness in the face and jaw muscle plus drooping eyelids.
- Cataracts, Cardiomyopathy, endocrine changes (insulin resistance.
- Speech and language difficulties.
- Developmental delay and learning problems.

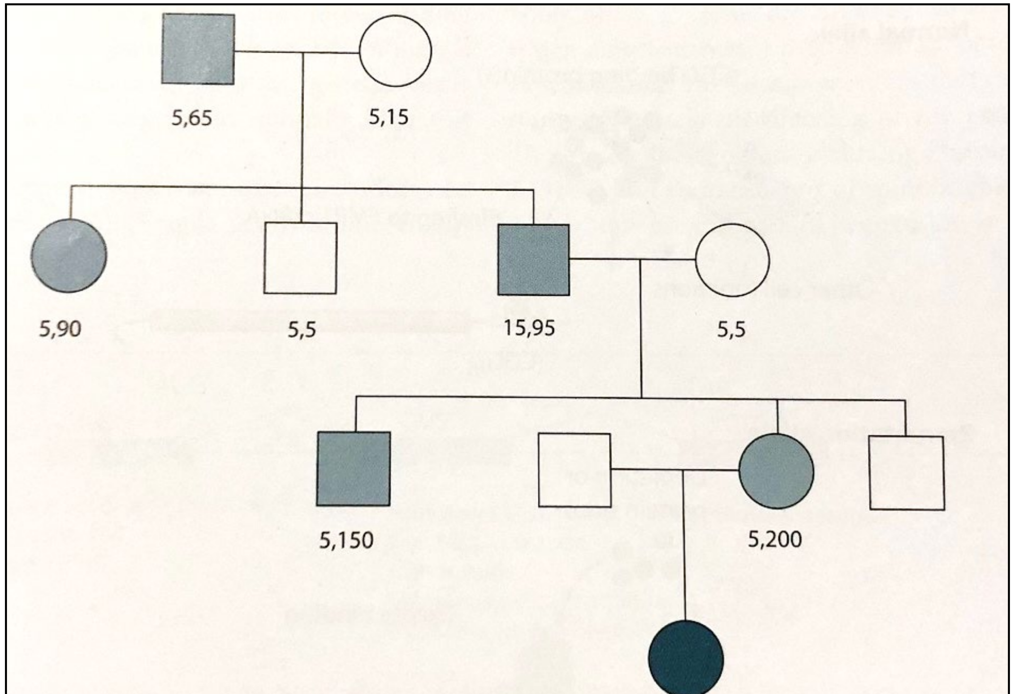


https://commons.wikimedia.org/wiki/File:Myotonic_dystrophy_patient.JPG

Myotonic Dystrophy:

3’ untranslated region

Pathogenic expansion



- Massive expansion along maternal lineage = severe anticipation.

Myotonic dystrophy

↓

RNA binding proteins, important for splicing, sequestered on the expanded repeat of DMPK mRNA

↓

Alteration of splicing for many mRNAs due to lack of RNA binding proteins = “***RNA toxicity***”

↓

Muscle pathology as *DMPK* is expressed in muscle!

Friedrich Ataxia:

- **Autosomal recessive disorder** - lack of muscle coordination during voluntary movements (neurodegeneration).

- Triple-nucleotide repeat in intron of **frataxin gene** = **altered chromatin structure** (formation of heterochromatin), **reduced transcription and loss of expression**.

- **No anticipation observed!**

- Frataxin involved in **mitochondrial iron metabolism** – required for heme cytochromes of ETC.

- Loss of expression results in iron overload in the mitochondria = **excessive oxidative damage.**

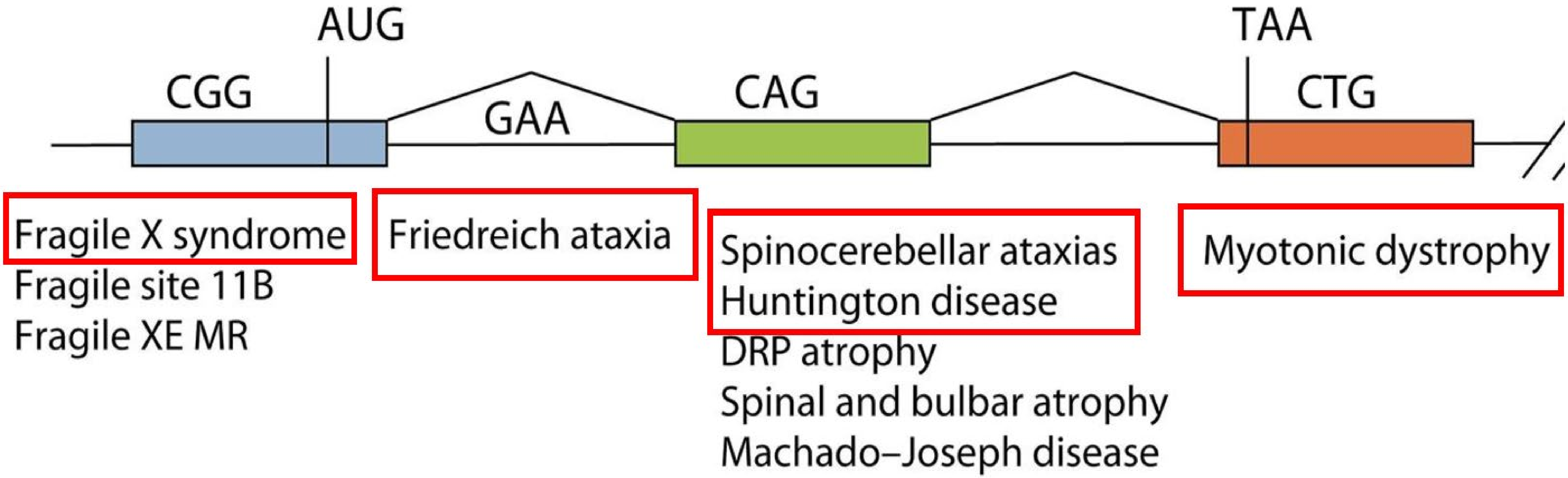
- Mitochondrial destruction responsible for observed symptoms.

Friedrich Ataxia:

Clinical Features:

- Onset 5 – 15 years of age (late onset 20 – 30 years of age).
- Symptoms: Difficulty walking and loss of tendon reflexes (ankles and knees)
  Slurring of speech (dysarthria), loss of vision and hearing.
  Scoliosis of the spine
  Diabetes
  Dysphagia (difficulty swallowing)
  Heart muscle degeneration
- Generally confined to a wheelchair 10 -20 years after onset.
- Heart disease/cardiomyopathy most common cause of death.

In Summary:



Human Genetics and Genomics, Fourth Edition. Bruce R. Korf and Mira B. Irons.

Thank you for listening!